

Introduction to Post Quantum Cryptography

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Part I - Basics of Quantum Computing

What is a Quantum Computer?

Common misconceptions:



A quantum computer is a computer that is faster at everything.



A quantum computation can brute force cryptographic keys

More accurate view:



A quantum computer is a computer that can solve certain specific problems much faster than any known classical algorithm, while offering no advantage for most everyday computation.

A Simple Intuition

Given a black-box function $f : \{0,1\}^n \rightarrow \{0,1\}$,

- There exists a unique x^* such that $f(x^*) = 0$
- For all other inputs x , $f(x) = 1$

Task: Find x such that $f(x) = 0$.

Note that the search space size: $N = 2^n$

Classical:

- Unstructured search
- $O(N)$ oracle queries in the worst case

Note: Think of x as a candidate AES key ($n = 128$), and $f(x)$ checks whether decryption “looks right”.

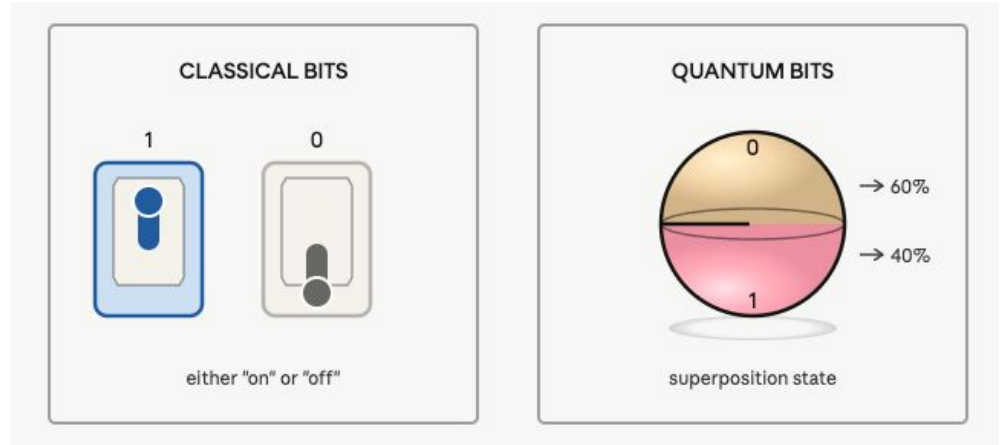
Quantum (Grover's algorithm):

- Uses superposition + interference
- $O(\sqrt{N})$ oracle queries

Bits vs. Qubit?

Classical computers

- Fundamental unit: **bit**
- Bit $\in \{0, 1\}$



Quantum computers

- Fundamental unit: **qubit**
- Qubit can be in a *quantum state* involving both $|0\rangle$ and $|1\rangle$



A qubit is **not** “both 0 and 1 at the same time” - it yields a definite value only when measured.

What is a Qubit?

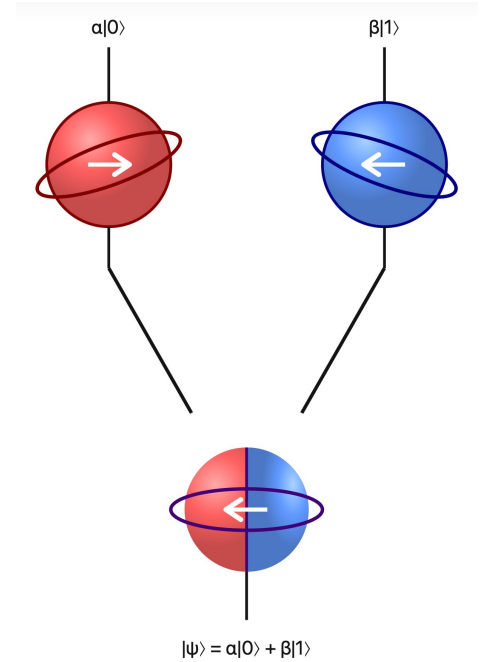
A qubit is described by a quantum state:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

where:

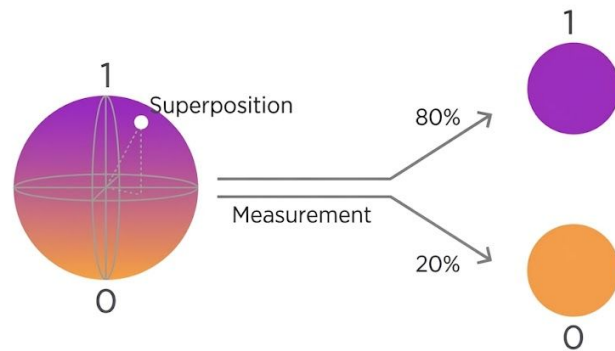
- $\alpha, \beta \in \mathbb{C}$ (Complex Numbers)
- $|\alpha|^2 + |\beta|^2 = 1$

α and β are probability amplitudes, not probabilities.



Measurement and Probability

- When we measure a qubit:
 - Outcome 0 occurs with probability $|\alpha|^2$
 - Outcome 1 occurs with probability $|\beta|^2$
- After measurement:
 - The qubit collapses to $|0\rangle$ or $|1\rangle$



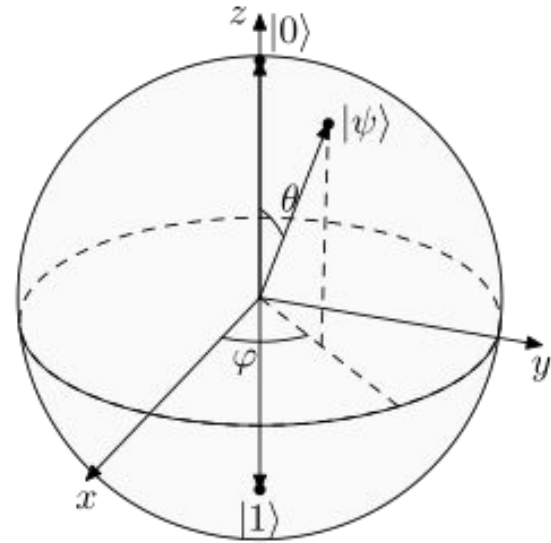
Quantum computers do **not** give access to all outcomes, measurement returns a single result.

Visualising a Qubit - The Bloch Sphere

- A single qubit state can be represented as a point on a sphere
- Different points correspond to different measurement outcome probabilities.

Key reference states

- $|0\rangle$ and $|1\rangle$ lie at opposite ends of the vertical axis (called the **computational basis**)



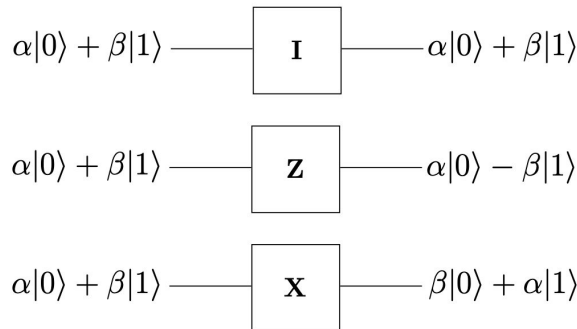
In addition to the computational basis $\{|0\rangle, |1\rangle\}$, the Bloch sphere contains other valid bases. For instance, $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$ form a basis along a different axis, as do $|i\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}$ and $|-i\rangle = (|0\rangle - i|1\rangle)/\sqrt{2}$. A basis specifies which measurement outcomes are possible and how likely each outcome is when the qubit is measured.

Single Qubit Gates

- Quantum gates transform the state of a qubit
- They are the basic operations used in quantum algorithms
- Single-qubit gates act as **rotations on the Bloch sphere**

X gate

- Maps $|0\rangle \leftrightarrow |1\rangle$
- Classical analogue of a NOT gate



Z gate

- Leaves $|0\rangle$ unchanged
- Changes the relative sign between $|0\rangle$ and $|1\rangle$
- Does not affect measurement probabilities in the $\{|0\rangle, |1\rangle\}$ basis

Single Qubit Gates - Continued

Hadamard gate

- Creates superposition
- Maps
 - $|0\rangle \rightarrow (|0\rangle + |1\rangle)/\sqrt{2}$
 - $|1\rangle \rightarrow (|0\rangle - |1\rangle)/\sqrt{2}$

$$|0\rangle \longrightarrow \boxed{H} \longrightarrow \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$|1\rangle \longrightarrow \boxed{H} \longrightarrow \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$



Applying H before measurement changes how the qubit is observed (measuring $H|\psi\rangle$ in $\{|0\rangle, |1\rangle\}$ is equivalent to measuring $|\psi\rangle$ in $\{|+\rangle, |-\rangle\}$)

Quantum Entanglement

Two Qubits and Entanglement

- **Entanglement** is a uniquely quantum correlation between qubits
- In an entangled state, the qubits **cannot be described independently**
- Measurement of one qubit **instantly determines** the state of the other

Example: simplest entangled state (Bell state)

$$|\Phi+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$$

- Measure the first qubit $\rightarrow$ outcome 0 $\rightarrow$ the second qubit is guaranteed 0
- Measure the first qubit $\rightarrow$ outcome 1 $\rightarrow$ the second qubit is guaranteed 1
- Individually, each qubit looks random (50% 0, 50% 1)
- Together, they are perfectly correlated

Superposition to Entanglement

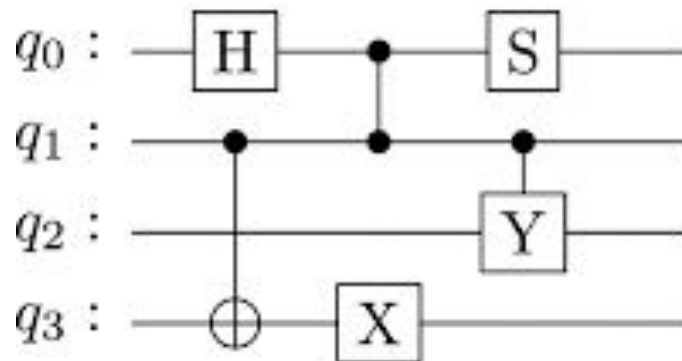
Why Entanglement Matters in Quantum Computing?

- Superposition alone allows multiple possibilities for a single qubit
- Entanglement allows correlations across qubits that have no classical equivalent
- These correlations are exploited in:
 - Quantum algorithms (e.g., Grover, Shor)
 - Quantum teleportation and communication
 - Quantum error correction

Quantum Gates

Concept

- Quantum algorithms are sequences of gates applied to qubits
- Represented visually as **circuit diagrams**
- Horizontal lines → qubits
- Boxes → gates
- Left-to-right → progression in time



Measurement and Quantum Interference

Quantum gates manipulate **probability amplitudes**, not probabilities

- Amplitudes are complex numbers and can:
 - Add together (**constructive interference**)
 - Cancel out (**destructive interference**)
- Measurement converts amplitudes into probabilities



Quantum algorithms carefully arrange gates so that wrong answers cancel out through interference, while the correct answer's amplitude is amplified and revealed by measurement.

From Quantum Circuits to Quantum Algorithms

- A quantum algorithm is a structured sequence of quantum gates
- Circuits are designed with a specific goal:
 - Prepare a useful quantum state
 - Transform it in a controlled way
 - Extract information by measurement

High-level workflow

- Initialize qubits (often into superposition)
 - Apply a carefully chosen gate sequence
- Measure to obtain the final result

Grover's Algorithm

Amplifying the Correct Answer

- **Objective:** Find the unique solution x^* to a black-box function $f(x)$ where $f(x^*) = 0$ in an unstructured search space of size $N = 2^n$.
- **The Quantum Speedup:** Classical brute-force requires $O(N)$ oracle queries in the worst case.
 - Grover's algorithm solves this in exactly $O(\sqrt{N})$ queries using **amplitude amplification**.

Grover's Algorithm - Workflow

1. State Initialization:

- Begin with n qubits initialized in the all-zero state.
- Apply a Hadamard gate to all qubits to create a uniform superposition of all possible N states, where every possible answer starts with the exact same initial probability amplitude.

2. The Grover Iteration (Repeated approximately π over four times the square root of N times):

- **Apply the Phase Oracle:** Flips the sign (relative phase) of the correct target state while leaving all other states unchanged.
- **Apply the Grover Diffusion Operator:** Performs a reflection about the average amplitude. This operation suppresses the amplitudes of the incorrect states and constructively amplifies the amplitude of the correct state.

Grover's Algorithm - Workflow

3. Measurement:

- Measure the final state in the standard computational basis.
- Because the amplitude of the correct state has been amplified close to 1, the correct solution is revealed with near-certainty.



Cryptographic Impact: Effectively halves the security margin of symmetric key lengths. To maintain a 128-bit security margin against a quantum adversary, systems must transition from AES-128 to AES-256.

Qubits and Quantum Error Correction

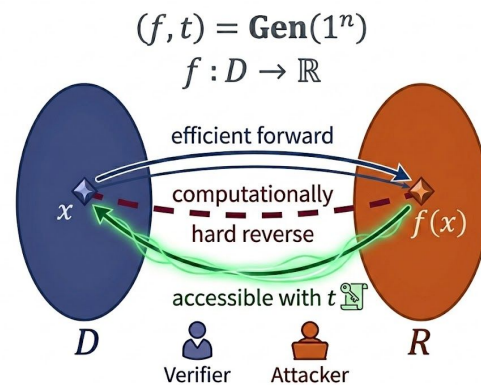
- Because individual physical qubits are incredibly fragile, a single reliable "logical" qubit is spread across an interconnected network of hundreds or thousands of physical qubits.
- Special circuits scan the system to catch error patterns without looking directly at the actual quantum data, preventing the delicate quantum information from collapsing.
- If the error rate of individual components is pushed below a specific mathematical threshold, adding more physical qubits will dramatically and continuously suppress overall system errors.
- Recent milestones, like Google's Willow demonstration, prove that building a reliable quantum computer has shifted from an unresolvable problem in physics to a practical manufacturing and engineering challenge.

Part II - Effect of Quantum Computing on Classical Cryptography

One-way functions and Trapdoors

- One-Way Functions & Trapdoors

- A **one-way function** is easy to compute but hard to invert
- Given an input x , computing $f(x)$ is efficient
- Given $f(x)$, finding x is computationally infeasible



- Trapdoor functions

- A **trapdoor** is secret information that makes inversion easy
- Without the trapdoor $\rightarrow$ inversion is hard
- With the trapdoor $\rightarrow$ inversion is efficient



Public-key cryptography relies on trapdoor functions. Public information allows encryption or verification. Secret information allows decryption or signing

Prime Factorization and the RSA Problem

- RSA Recap

- Pick two large prime numbers p and q
- Compute $n = pq$
- Compute $\Phi(n) = (p-1)(q-1)$
- Select public key e , s.t. $\gcd(e, \Phi(n))=1$
- Compute the private key d , s.t. $d = e^{-1} \bmod(\Phi(n))$
- Delete p , q and $\Phi(n)$
- (e, n) is the public key
- (d, n) is the private key

Encryption $c = m^e \bmod n$ Decryption $m = c^d \bmod n$

- Security of RSA

- RSA Inversion Problem (finding the e^{th} root of a given number)

i.e., given c can you find m which equivalent to finding the e^{th} root of m , modulus n

- Prime factorisation problem

i.e., if you can factorise n and find p and q , then you can find $\Phi(n)$. Then using e , which is publicly known and $\Phi(n)$ you can find d (Using the extended Euclidean algorithm)

Implications of Breaking RSA

- No efficient classical algorithm exists for factoring large integers.
 - The best algorithm is ~ **sub-exponential time** far worse than **$O(N^2)$** .
- Using **Shor's algorithm**, a quantum computer can factor integers in **polynomial time** (specifically, **$O((\log N)^3)$**)
- For RSA-2048, Shor's algorithm requires ~**4099 logical qubits**.
 - Needs millions of **physical** qubits, far beyond current hardware.
- Currently, the only known way to efficiently solve RSA inversion is by factoring n . We **do not know of any other classical or quantum algorithm** that can invert RSA without factoring.

Part III - NIST PQC Standards

Why a Standardization Process

- Shor's algorithm threatens RSA, DSA, ECDSA, DH, ECDH — nearly all deployed public-key crypto.
- The community already had many "quantum-safe" candidates — [lattices](#), [codes](#), [hashes](#), [isogenies](#), [multivariate](#) — but:
 - No agreed parameters, no agreed security levels.
 - No common API, no interoperability.
 - "Harvest-now, decrypt-later" makes the timeline urgent. - Detour

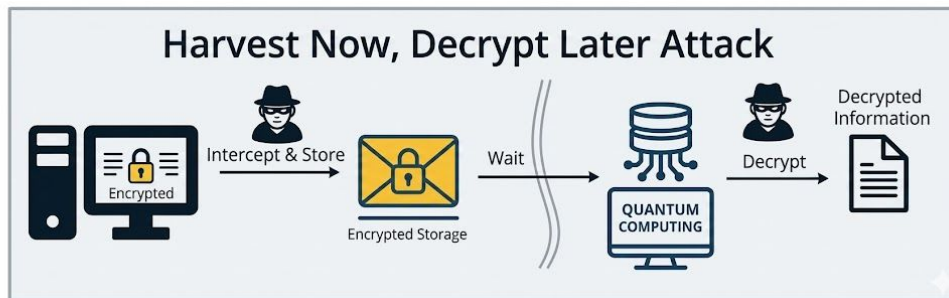
Real-World Example: TLS "Harvest Now, Decrypt Later"

Typical TLS 1.3 Session Today (2026):

- Client and server perform **ECDHE** key exchange
- Client sends ephemeral public key: g^a (where a is secret)
- Server sends ephemeral public key: g^b (where b is secret)
- Both compute shared secret: $g^{ab} \rightarrow$ derive AES session key
- All traffic encrypted with AES-256-GCM
- Attacker sees g^a and g^b on the wire
- Cannot compute a from g^a (discrete log is hard)
- Cannot compute g^{ab} without a or b
- Forward secrecy protects past sessions

Quantum Threat (Future ~2030s):

- Harvest Now (2026): Adversary records and stores:
 - Captured g^a, g^b
 - All encrypted traffic (ciphertext)
- Decrypt Later (~2030s): With quantum computer:
 - Use Shor's algorithm to solve discrete log
 - Compute a from g^a and p
 - Compute $g^{ab} \rightarrow$ recover session key
 - Decrypt all stored traffic



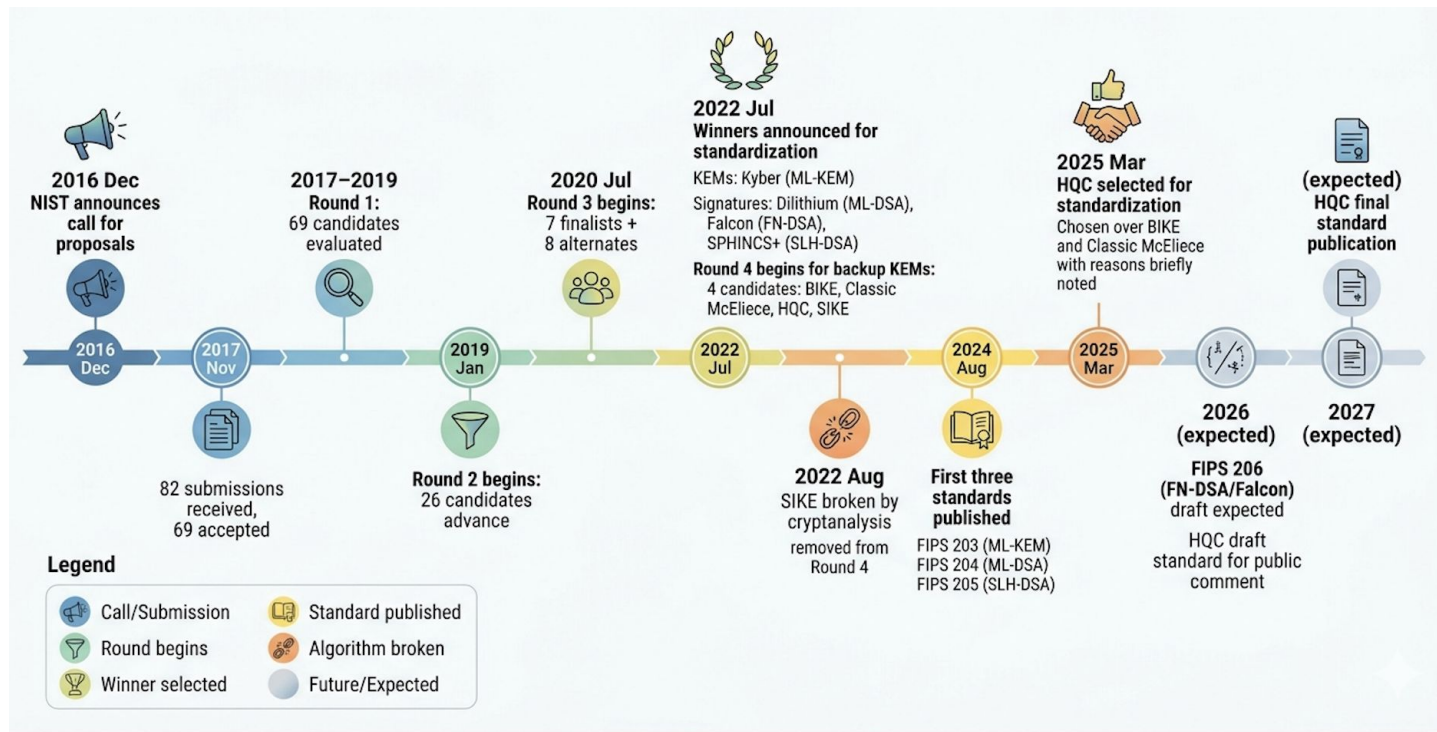
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A trusted body needed to evaluate candidates, pick winners, and publish reference parameters that vendors can implement against.

High-Level Timeline



Final NIST Standards - As FIPS Documents

Published in their final form on **13 August 2024** (one still in draft):

FIPS	NIST name	Originally called	Purpose	Family
FIPS 203	ML-KEM	CRYSTALS-Kyber	Key Encapsulation	Lattice (Module-LWE)
FIPS 204	ML-DSA	CRYSTALS-Dilithium	Digital Signature	Lattice (Module-LWE/SIS)
FIPS 205	SLH-DSA	SPHINCS+	Digital Signature	Hash-based
FIPS 206 <i>(draft)</i>	FN-DSA	FALCON	Digital Signature	Lattice (NTRU)
<i>(separate)</i>	<i>HQC</i>	HQC	Backup KEM (Mar 2025)	Code-based



NIST renamed the schemes when they became standards. The new names describe the construction, not the team. Both names will be in use for years

Decoding the New Names

The NIST naming convention is functional:

- **ML-** = **M**odule-**L**attice (Module-LWE based)
 - **SLH-** = **S**tateless **H**ash-based
 - **FN-** = **F**ourier-**N**TRU (NTRU + FFT sampling)
 - **-KEM** = Key Encapsulation Mechanism
 - **-DSA** = Digital Signature Algorithm
-
- **ML-KEM** = Module-Lattice based Key Encapsulation Mechanism
 - **ML-DSA** = Module-Lattice based Digital Signature Algorithm
 - **SLH-DSA** = Stateless Hash-based Digital Signature Algorithm
 - **FN-DSA** = FFT-over-NTRU Digital Signature Algorithm

Terminology Shift — "Key Exchange" → "Key Encapsulation"

Classical Diffie–Hellman (key exchange)

- Alice and Bob each publish g^a and g^b .
- Both independently compute the same shared secret g^{ab} .
- Genuinely symmetric — an *exchange*.

Post-Quantum KEM

- Alice publishes a public key. Bob has no published key.
- Bob picks a random session key K , encapsulates it under Alice's p_k to produce a ciphertext c .
- Alice decapsulates c with her secret key to recover K .
- Closer to public-key encryption of a random session key than a symmetric exchange.



Lattice schemes do not give a clean commutative group operation like $g^{ab} = g^{ba}$.
The natural primitive is "encrypt a random session key" — hence KEM.

Underlying Hard Problems

Just as classical PKC rests on DLP and factoring, the standards rest on

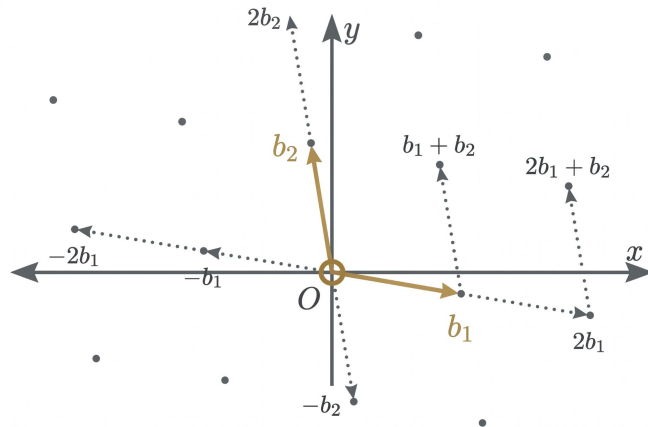
Standard	Underlying problem	Classical analogue (vibes only)
ML-KEM, ML-DSA	Module-LWE / Module-SIS	"Lattice version" of DLP — solving noisy linear equations over $\mathbb{Z}_q$
FN-DSA	Short vectors in NTRU lattice	Like DLP but on a structured lattice rather than a cyclic group
SLH-DSA	Hash one-wayness / collision resistance	No number theory — pure symmetric security
HQC	Decoding random linear codes	Different world — coding theory



Lattice and code problems are believed to resist Shor's algorithm because they don't reduce to the Hidden Subgroup Problem over an Abelian group — the structure Shor exploits for factoring and DLP.

Lattices and ML-KEM

A lattice is the set of all integer combinations of a set of selected vectors



- **SVP** (Shortest Vector): find the shortest nonzero lattice vector.
- **CVP** (Closest Vector): given target t , find the closest lattice point.
- **LWE**: given many noisy linear samples $(a_i, \langle a_i, s \rangle + e_i)$, recover s — reduces to lattice problems.

Easy in 2D. Hard in hundreds of dimensions — even for a quantum computer.

ML-KEM How it works

Based on Module Learning With Errors (MLWE).

KeyGen()

- Sample a public matrix A over $\mathbb{Z}_q$.
- Sample small secret s and small noise e .
- Compute $t = A \cdot s + e$.
- $pk = (A, t)$, $sk = s$

Encaps(pk)

- Sample small r, e_1, e_2 and a random message m .
- $u = A^T \cdot r + e_1$
- $v = t^T \cdot r + e_2 + \lceil q/2 \rceil \cdot m$
- $c = (u, v)$, $K = \text{Hash}(m)$.

Decaps(sk, c)

- Compute $v - s^T \cdot u \approx \lceil q/2 \rceil \cdot m$.
- The noise is small enough that m can be rounded out.
- Recover m , return $K = \text{Hash}(m)$



The noise e is what makes the public key hard to invert, but small enough that the secret s can cancel it during decryption.

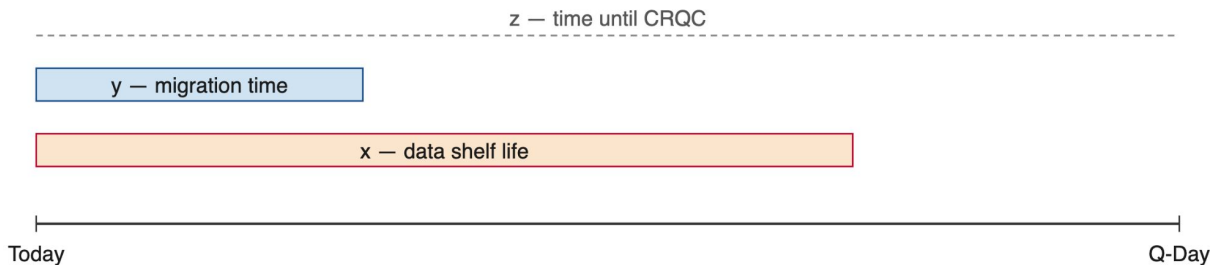
Part IV - Timeline, Adoptions and the Hybrid Question

Why the Timeline Matters - Mosca's Inequality

Three numbers describe your exposure:

- **x** - how long your data must stay confidential (shelf life)
- **y** - how long it takes you to migrate to PQC
- **z** - how long until a cryptographically relevant quantum computer (CRQC) exists

If $x + y > z$, you are already too late



Data encrypted today with RSA/ECC may be captured today and decrypted in 10–20 years — the "harvest now, decrypt later" threat. Long-lived secrets (medical records, state secrets, IP) are at risk right now, not when Q-Day arrives.

When will be the Q-Day?

- "Q-Day" / "Y2Q" — the day a quantum computer can break RSA-2048 / ECDSA-P256.
 - CRQC (Cryptographically Relevant Quantum Computer)
- Estimates have been shrinking with every survey:

Year of survey	"Likely within 10 years"
2015	~15%
2019	~22%
2022	~28%
2024	>35%

- Some researchers now consider **2030–2035** plausible.
- Also reflected in various government initiatives globally.

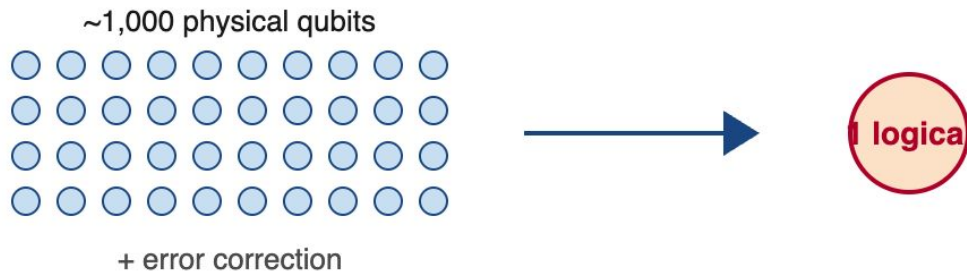
Recent Quantum Computing Breakthroughs

Date	Milestone
Dec-23	IBM Condor — 1,121 superconducting qubits (raw scale)
Dec-23	IBM Heron r1 — 133 qubits, much lower error rate
Dec-24	Google Willow — 105 qubits; demonstrated below-threshold quantum error correction (errors halve as code distance grows)
Late 2024	Atom Computing — 1,180+ neutral-atom qubits
Feb-25	Microsoft Majorana 1 — first claimed topological-qubit processor (8 qubits; scientific claims contested)
Feb-25	AWS Ocelot — 9-qubit cat-qubit prototype with hardware-level error suppression
2025	IBM roadmap → "Starling" (200 logical qubits by 2029)



What matters is not raw qubit count but error correction. Willow's below-threshold demonstration was the long-awaited signal that fault-tolerant scaling is real, not hypothetical.

Physical vs. Logical Qubits



Breaking RSA-2048 needs ~4,000 logical qubits

→ on the order of

Millions of Physical Qubits

- Until 2024, this seemed an impossibly distant goal.
- Willow (Dec 2024) showed that adding more qubits actually *reduces* the logical error rate.
- The bottleneck is no longer "in principle unsolvable" — it is now a scaling problem.

National Roadmaps - United States

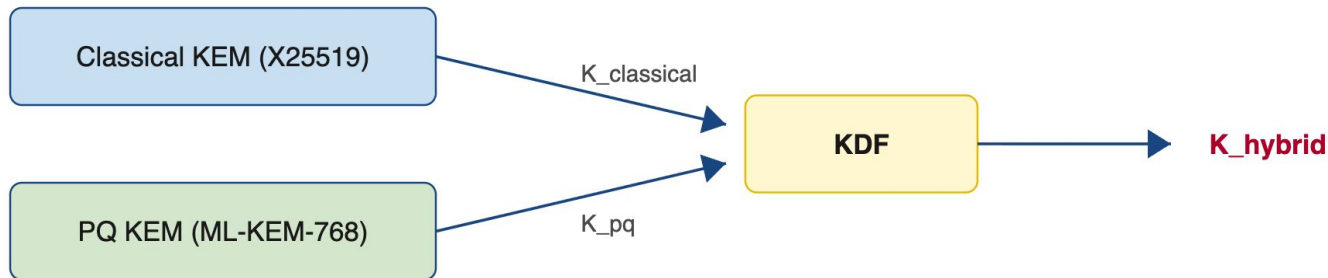
- NSM-10 (May 2022) directs all US federal systems toward PQC.
- NSA CNSA 2.0 (September 2022) — schedule for National Security Systems:

Use case	Prefer PQC by	Exclusive PQC by
Software & firmware signing	2025	2030
Networking equipment	2026	2030
Web browsers, servers, cloud	2025	2033
Operating systems	2027	2033
Custom / niche applications	2030	2033
All NSS systems	—	2035

- Summary - Deprecate RSA-2048/ECC P256 by 2030 and Disallow all RSA and ECC by 2035. Similar guidelines from UK, Germany, Australia etc. Some are even more tighter. Important note hybrid (more on this later).

Hybrid Cryptography

- Run a classical KEM (e.g. X25519) **and** a PQ KEM (e.g. ML-KEM-768). Combine the two shared secrets:



- An attacker must break both to recover K_{hybrid} .
- Used in TLS 1.3 (X25519MLKEM768), Signal's PQXDH, Apple iMessage PQ3, and most production deployments today.



If only PQC falls: classical still protects you. If only classical falls (eventually via Shor): PQC still protects you. You only lose if both fall.

Current Adoption Levels of PQC

- **Web & Browser Integration:** Over 38% of global HTTPS traffic is now post-quantum secured via hybrid X25519+ML-KEM across Google Chrome and Cloudflare edge networks.
- **Hyperscalers:** AWS (Key Management Service) and Google Cloud have standardized ML-KEM (FIPS 203) for internal data transit to neutralize "harvest now, decrypt later" threats.
- **Secure Messaging Maturity:** Signal (PQXDH protocol) and Apple iMessage (PQ3 architecture) lead with quantum security, utilizing high-frequency re-keying for long-term message secrecy.
- **Enterprise Admin Connectivity:** OpenSSH (9.0+) and corporate VPN gateways have shifted to hybrid PQC by default, protecting administrative tunnels from quantum eavesdropping.
- **Blockchain Migration Hurdles:** While standard web and transport-layer traffic have pivoted rapidly, the immutability and decentralized nature of public ledgers (like Bitcoin and Ethereum) remain the primary remaining hurdle for quantum-safe signature migration.

What to do Today?

- **Inventory** your cryptographic dependencies (libraries, certs, HSMs, code signing).
- **Enable hybrid PQ TLS** on internet-facing endpoints — most modern stacks support it.
- **Prioritise long-shelf-life data** (medical, IP, state secrets, root keys, code signing) — these are exposed to harvest now, decrypt later.
- **Track vendor roadmaps and watch national guidance**
- Test now - staging deployments of ML-KEM, ML-DSA, SLH-DSA today are cheap insurance.

Summary

- Timelines are shrinking. Willow (Dec 2024), Majorana 1 and Ocelot (Feb 2025) made fault-tolerant quantum computing concrete, not hypothetical.
- Governments have set deadlines — broadly 2030 for high-value data, 2035 for the long tail. Several were tightened in 2024–2025.
- The web has already moved — over half of TLS connections were PQ-protected by end of 2025.
- Hybrid PQC is the dominant transition strategy — recommended (or required) by BSI, ANSSI, NCSC; deployed by Chrome, Signal, Apple, Zoom. Pure PQC is preferred by NSA for NSS and is the long-term destination.
- Blockchains and HSM-backed systems are the hardest cases — start now.